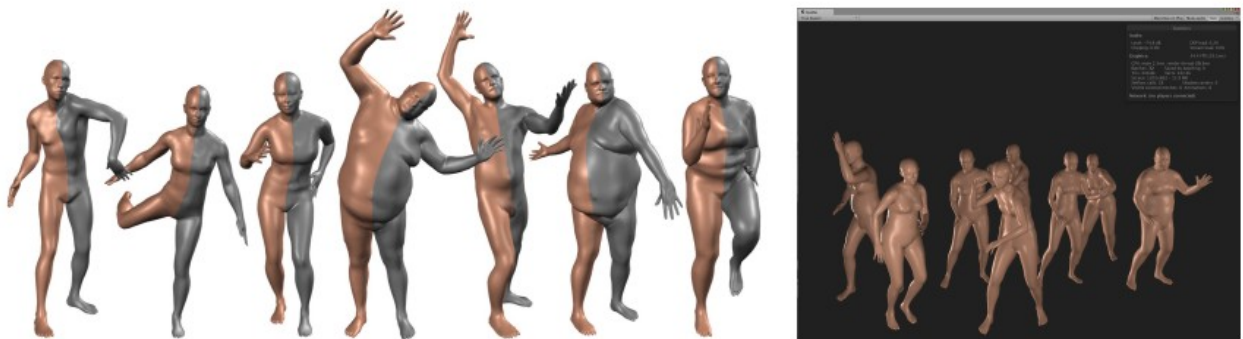


Neural Localizer Fields for Continuous 3D Human Pose and Shape Estimation

The research on "*Neural Localizer Fields for Continuous 3D Human Pose and Shape Estimation*¹" represents an innovative leap in the field of 3D human modeling, combining human pose and shape estimation into a continuous representation framework. The work focuses on overcoming limitations seen in traditional methods, where discrete joint-based models and rigid parametric shape representations are often inadequate for real-world applications.

1. Motivation and Problem Statement

- Human pose and shape estimation has traditionally been limited by the discrete nature of body models, which are either based on a set of keypoints (e.g., joint positions) or predefined body shapes (e.g., *SMPL*²³ model). These methods tend to miss the fine-grained details of human bodies, especially when dealing with diverse body shapes and dynamic poses.
- The motivation behind this research is to develop a more flexible, accurate, and continuous representation of both human pose and shape, which can better adapt to real-world variations and offer smoother modeling of human body movements.



2. Core Contributions

- **Neural Localizer Fields:** The proposed approach introduces "*localizer fields*" as a means of encoding continuous 3D space. These fields allow for precise localization of body parts and more accurate body shape and pose estimation, especially when compared to traditional discrete representations. The key idea here is the use of neural networks that map input data (e.g., 2D images or depth scans) into these continuous fields.
- **End-to-End Model:** The research suggests a neural network that can directly predict the 3D pose and shape of the human body from input data, eliminating the need for intermediate

1 [NeurIPS'24 paper](#)

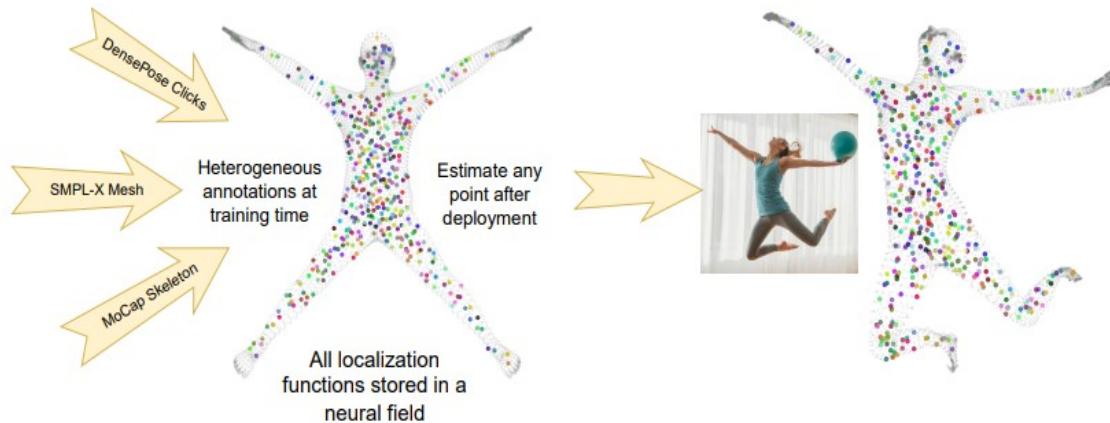
2 [PDF of the SMPL model](#)

3 [A Skinned Multi-Person Linear Mode](#)

predefined models. This end-to-end framework offers significant flexibility in capturing diverse poses and body shapes.

3. Technical Approaches

- **Continuous 3D Representation:** One of the main technical challenges addressed is representing the 3D body model continuously, rather than relying on discrete voxel grids or mesh-based approaches. Continuous 3D representations, such as neural radiance fields (*NeRF*) or other implicit functions, can provide a smoother and more detailed model of the human body.



- **Method:** The method consists of three main parts: (1) a *point localizer network (PLN)* for localizing 3D points from images, (2) a *neural localizer field (NLF)* to control what point the *PLN* should predict, and (3) an efficient body model fitting algorithm to create a parametric representation for a set of points that were localized nonparametrically. This complete approach is called *NLF*. This modular structure establishes clear interfaces and can enable independent improvements on the parts.

- **Point Localizer Network (PLN)**

- **The Point Localizer Network (PLN)** is central to the architecture for localizing 3D points from RGB image crops. This section describes the core mechanism for localizing a point, which is later extended to handle arbitrary points in the subsequent part of the architecture. An image crop (I) of dimensions $H \times W \times 3$ (height, width, RGB channels) is fed into the system.

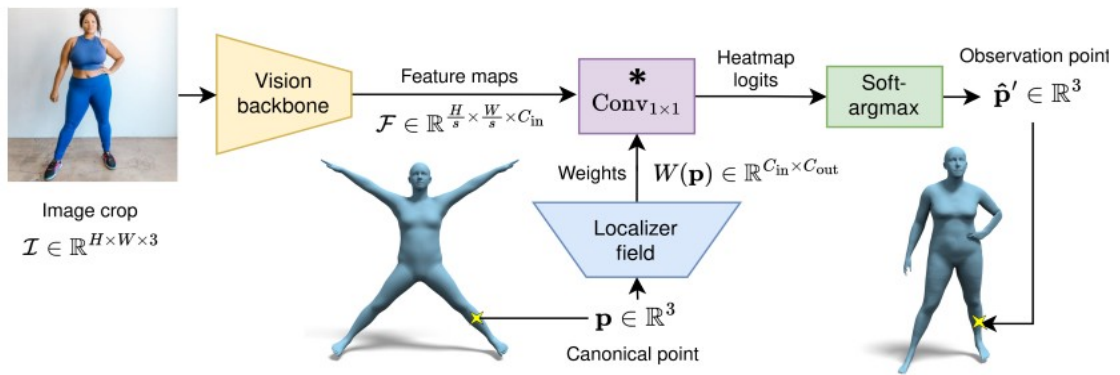
- **Neural Localizer Field (NLF)**

- The core idea behind the **Neural Localizer Field (NLF)** is to go beyond predicting a fixed, predetermined set of body points and instead enable the localization of **all possible points** in the body using a **universal model**. This is achieved by learning a **continuous field of localizer functions** denoted as Ψ , where each function f_p localizes a specific point p in the **canonical human volume** $p \in \mathbb{R}^3$.

- **Efficient Body Model Fitting in Post-Processing**

- While nonparametric point localization provides high accuracy in fitting human body parts from images, a **parametric representation**—such as using joint rotations θ and shape vector β —can be more compact and offer better disentanglement of **pose** and **shape**. Therefore, the authors introduce an efficient post-processing algorithm to **fit a parametric model** (like SMPL) to the nonparametrically predicted 3D points.

The architecture overview is depicted in next diagram:



The diagram represents a pipeline for continuous 3D human pose and shape estimation using a neural localizer field. Here's an analysis of its components:

- **Input Image :**

- An image crop I of dimensions $H \times W \times 3$ (height, width, RGB channels) is fed into the system.

- **Vision Backbone:**

- The image is passed through a vision backbone (e.g., a convolutional neural network) to extract feature maps

$$\mathcal{F} \in \mathbb{R}^{\frac{H}{s} \times \frac{W}{s} \times C_{in}}$$

where s is the spatial downsampling factor, and C_{in} is the number of feature channels.

- **Convolutional Layer (Conv_{1x1}):**

- A 1×1 convolution is applied to produce heatmap logits, potentially reducing the channel dimension to C_{out} . These logits represent the likelihood of key points in the feature space.

- **Soft-argmax:**
 - The heatmap logits are passed through a soft-argmax operation to generate predicted 3D observation points:

$$\hat{\mathbf{p}}' \in \mathbb{R}^3$$

- **Localizer Field:**
 - A crucial component is the **Localizer Field**, which maps a canonical 3D point ($p \in \mathbb{R}^3$) to weight tensors $W(p) \in \mathbb{R}^{C_{in} \times C_{out}}$
 - These weights are used to adaptively fuse feature maps to predict the 3D locations corresponding to canonical points.
- **Canonical and Observation Points:**
 - The diagram visualizes canonical points (e.g., a star on a posed human model) and the estimated observation points on the reconstructed shape.
 - The canonical points are mapped to observation points $\hat{\mathbf{p}}'$ using the learned localization field and other network components.

4. Advantages of the Approach

- **Improved Accuracy:** The continuous representation provides higher fidelity in capturing body details, improving the accuracy of both pose and shape estimation.
- **Smoother Transitions:** Unlike discrete models, which can produce jagged or unnatural transitions between body parts, continuous fields allow for smoother and more realistic representations of the human body in motion.
- **Greater Flexibility:** The method is expected to be more adaptable, able to handle a broader range of body types and movements, especially in complex scenarios like occlusions or highly dynamic actions.

5. Challenges

- **Computational Demands:** Training a model that operates on continuous 3D representations is computationally expensive, requiring significant resources in terms of processing power and memory. Optimizing such a model for real-time use, particularly in applications like AR/VR or robotics, could be a challenge.
- **Data Needs:** To train the model effectively, a large amount of annotated 3D data is typically required. Gathering such datasets, especially for diverse and realistic human shapes, can be time-consuming and costly.
- **Generalization to New Scenarios:** Ensuring that the model generalizes well to new body types, movement styles, and environments is critical. Variability in human body shapes and the complexity of real-world motion are challenges that need to be addressed to make the model robust.

6. Applications

- **Motion Capture:** This method can revolutionize motion capture technology, providing a more detailed and accurate 3D model of human movement. It can be used in film, animation, and game development to create lifelike character animations.
- **Robotics and Assistive Devices:** Robots or assistive devices that need to interact with humans can benefit from this more accurate understanding of human pose and shape. It can help robots learn human motions for better interaction or design personalized movements.
- **Augmented Reality (AR) and Virtual Reality (VR):** In AR/VR, the ability to create highly accurate and detailed human avatars is critical. Continuous pose and shape estimation can make virtual avatars appear more lifelike, improving user experience.
- **Sports Science and Biomechanics:** Analyzing human movement in detail can aid in sports science, helping to optimize performance, prevent injuries, and study biomechanics.

7. Relation to Existing Methods

- **SMPL and Parametric Models:** While SMPL and other parametric models provide a good foundation for human shape estimation, they are limited by their rigid structure and inability to model continuous variations in shape. The neural localizer fields approach aims to overcome these limitations by providing a more flexible and detailed model.
- **Implicit Neural Representations (e.g., NeRF):** The research draws from concepts used in neural radiance fields (NeRF), which represent 3D scenes continuously using neural networks. These methods have been successful in other domains and are being adapted to human pose and shape estimation.
- **Graph Neural Networks (GNNs):** Earlier methods in pose estimation often relied on GNNs to capture the spatial relationships between body joints. This research could combine GNNs with continuous 3D representations to enhance pose estimation further.

8. Future Directions

- **Real-Time Estimation:** One of the logical next steps is making the method efficient enough for real-time applications. Optimizing the network for faster inference could expand its use in interactive environments like video games or robotics.
- **Fusion of Multiview Data:** The model could potentially integrate multiple sensor inputs, such as depth cameras or inertial measurement units (IMUs), to improve accuracy and robustness, particularly in cases of occlusion or complex environments.
- **Personalization:** As this research develops, creating models that can be personalized to different body types, clothing, and individual movement styles could further improve its real-world application.

Conclusion

The approach of using neural localizer fields for continuous 3D human pose and shape estimation offers significant advantages over traditional methods, providing more detailed, flexible, and accurate models of the human body. By overcoming the limitations of discrete models and rigid parametric representations, it opens up new possibilities for motion capture, robotics, VR/AR, and sports science. While the approach is promising, challenges such as computational demands, data requirements, and generalization remain to be addressed. Future advancements may focus on making the model real-time capable and applicable to a broader range of real-world scenarios.

Test the demo and show the results:

The models for PyTorch and TensorFlow are available for non-commercial research use in Releases, and examples of usage are given in `demo.ipynb`, but to get them to work for me I had to modify the code as follows:

To work for PyTorch version in the “demo.ipynb”, I changed the code like this and saved in “TF.txt” file:

```
import torch
import torchvision # Must import this for the model to load without error

# Create models directory if it doesn't exist and download the model
!mkdir -p models
!wget -q -O models/nlf_1_multi.torchscript https://bit.ly/nlf_1_pt # Loads the
TorchScript model

# Load the TorchScript model to the GPU and set it to evaluation mode
model = torch.jit.load('models/nlf_1_multi.torchscript').cuda().eval()

# Import files.upload to upload image
from google.colab import files

# Upload the image file (user will upload the image interactively)
uploaded = files.upload()

# Assuming the user uploads 'example_image.jpg', we read it using torchvision
image_path = 'example_image.jpg'

# Read the image and move it to the GPU
image = torchvision.io.read_image(image_path).cuda()

# Ensure the image has the correct shape and dimensions
print(f"Image shape: {image.shape}") # Should be [C, H, W] for a 3-channel image

# Prepare the image by adding a batch dimension
frame_batch = image.unsqueeze(0)

# Perform inference using the model
with torch.inference_mode(), torch.device('cuda'):
    pred = model.detect_smpl_batched(frame_batch)

# SMPL Parametric predictions
print("Pose parameters:", pred['pose'])
print("Shape parameters:", pred['betas'])
print("Translation parameters:", pred['trans'])
print("3D Joints:", pred['joints3d'])
print("3D Vertices:", pred['vertices3d'])
print("2D Joints:", pred['joints2d'])
```

```

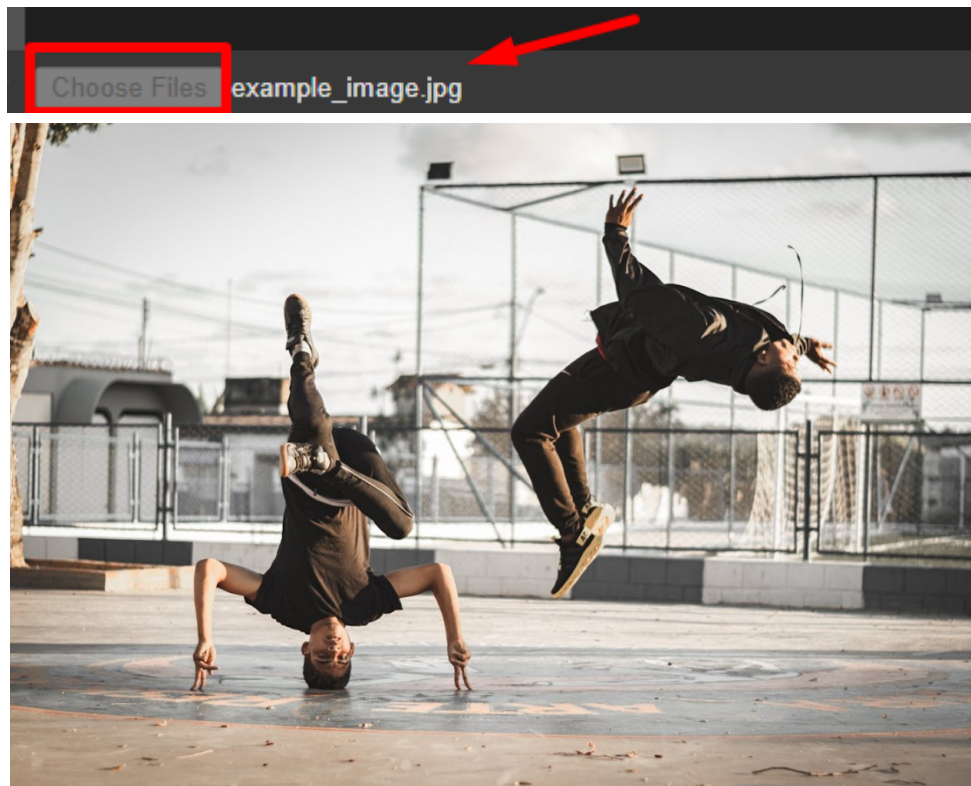
print("2D Vertices:", pred['vertices2d'])

# Nonparametric joints and vertices
print("Nonparametric 3D Joints:", pred['joints3d_nonparam'])
print("Nonparametric 3D Vertices:", pred['vertices3d_nonparam'])
print("Nonparametric 2D Joints:", pred['joints2d_nonparam'])
print("Nonparametric 2D Vertices:", pred['vertices2d_nonparam'])

# Joint and vertex uncertainties
print("Joint Uncertainties:", pred['joint_uncertainties'])
print("Vertex Uncertainties:", pred['vertex_uncertainties'])

```

import the image “example_image.jpg” from the project directory:



and I saved the output in the “Torch_Result.txt” file.

For “Tensorflow” version works correct and the result is saved in “Tensorflow_Result.txt” file.

References:

[Project page](#)

[Project code](#)

[PDF of the SMPL model](#)

[A Skinned Multi-Person Linear Model](#)

[PosePile - a large collection of dataset processing](#)